

Deploying with Argo CD Across Multiple Clusters 🌐

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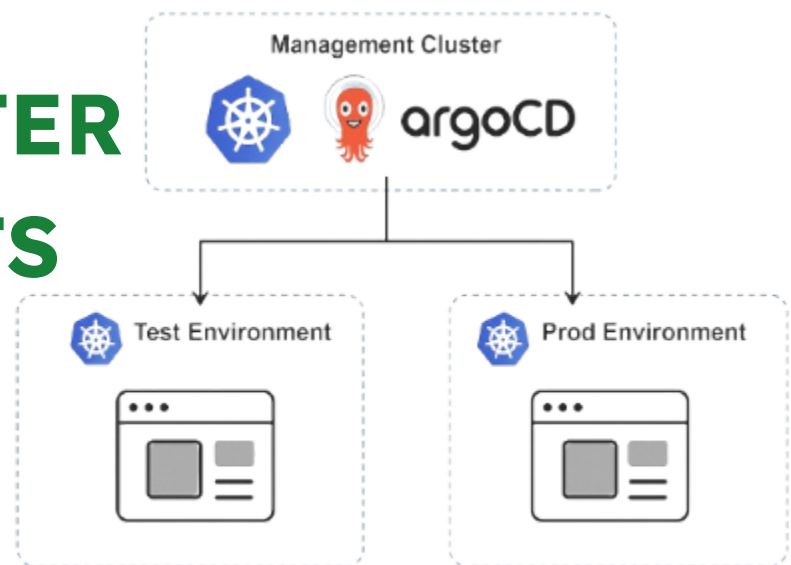
What is Argo CD? 🤔

Argo CD is a **GitOps** tool.

That means it looks at a Git repo 📁 and makes sure your Kubernetes clusters look the same as the files in Git. It keeps watching and syncing the clusters to match.

Think of it like a robot 🤖 that keeps all your clusters in line with your Git code — like magic ✨!

MULTI-CLUSTER DEPLOYMENTS With ArgoCD



Why Use ArgoCD With Multiple Clusters? 🕸

In real companies, you don't have just one cluster. You might have:

- 🧪 **Test cluster** (for testing code)
- 🧪 **Staging cluster** (for checking before going live)

- 🚀 **Production cluster** (for real users)

They use **Argo CD** to control all these clusters from **one place** 🎮. It saves time and reduces mistakes!

The Setup ⚙️

Here's what you need to do to use Argo CD with multiple clusters:

1. 🛠️ Set up Argo CD on One Cluster (Called "Hub")

Install Argo CD on one cluster — maybe the test or a "management" cluster.
This is the **main control center** (called the hub).

```
kubectl create namespace argocd
kubectl apply -n argocd -f
https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
```

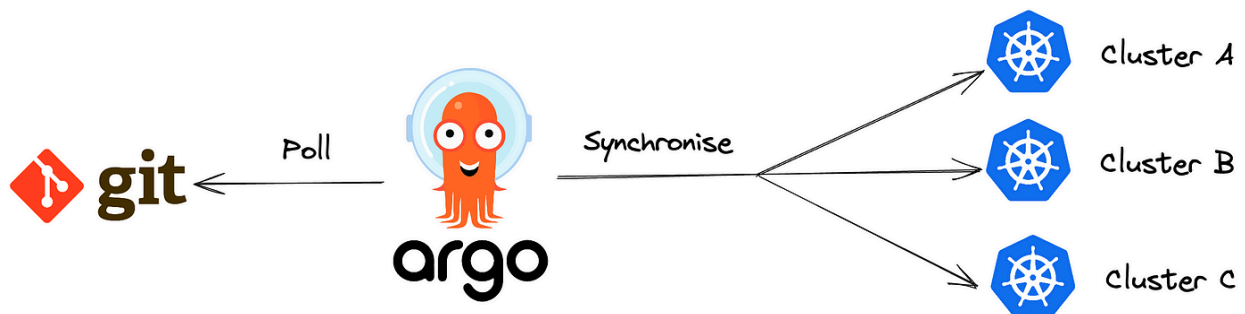
2. 🔗 Connect Other Clusters (Called "Spokes")

Use `argocd cluster add` to connect other clusters (like staging or production).
These clusters will get updates from the main Argo CD server.

```
argocd cluster add <context-name-of-other-cluster>
```

You must run this from the Argo CD CLI inside the hub cluster.

✅ After this, Argo CD knows about your other clusters.



Dealing with Private Docker Images (ECR)

If your Kubernetes deployments use images from AWS ECR (private), you need to give your clusters permission to pull those images.

Use a Kubernetes Secret for Docker Login

You can make a `docker-registry` secret and tell Argo CD to use it.

```
apiVersion: v1
kind: Secret
metadata:
  name: my-ecr-secret
  namespace: default
type: kubernetes.io/dockerconfigjson
data:
  .dockerconfigjson: <base64 string here>
```

Then, in your deployment YAML:

```
imagePullSecrets:
- name: my-ecr-secret
```

👉 This way, **your other clusters** can pull images too.

Syncing Apps to Different Clusters

When you define your application in Argo CD, you tell it which cluster to deploy to.

Here's a simple `Application` YAML:

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: my-app
  namespace: argocd
```

```
spec:
  destination:
    server: https://<other-cluster-api>
    namespace: default
  source:
    repoURL: https://github.com/your-org/your-repo
    path: apps/my-app
    targetRevision: main
  project: default
  syncPolicy:
    automated: {}
```

Argo CD reads this, goes to that cluster, and deploys the app there! 🚀

Real-Life Example: Org with 3 Clusters 🧠

Let's say your company has:

- **Dev Cluster** in us-east-1
- **Stage Cluster** in us-west-1
- **Prod Cluster** in eu-central-1

You:

1. Deploy Argo CD on **Dev**
2. Add **Stage** and **Prod** using `argocd cluster add`
3. Create apps in Argo CD with `destination.server` pointing to the right cluster

4. Boom 🌟 — Argo CD deploys to all 3 clusters using Git!
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Tips & Notes 🧠

- 🛡️ Use **IAM Roles** or **Secrets** to allow ECR access in each cluster
 - 📁 Keep each environment (dev, stage, prod) in its own folder or Git branch
 - 🔄 Use **Auto-Sync** for zero manual deployment
 - 🧪 Test everything in **Dev first**, then push to Prod
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